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Lab Session 5: Conditional Statements

C/C++

```
if marks >= 50:
```

Lab Session 5: Conditional Statements

Lab Manual: Conditional Statements in Python

LAB MANUAL: CONDITIONAL STATEMENTS IN PYTHON

Course/Subject: Python Programming

Topic: Conditional Statements

Level: Beginner

Lab Type: Practical Programming Lab

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LAB OBJECTIVES

- By the end of this lab, students will be able to:
- Understand the concept of decision-making in Python.
- Use if statements to execute code based on a condition.
- Use if-else statements to handle two possible outcomes.
- Use if-elif-else statements to handle multiple conditions.
- Use nested if statements for decision-making inside another condition.
- Apply comparison operators and logical operators in conditions.
- Solve real-world problems using conditional statements.
- Write Python programs using proper indentation and syntax.

BACKGROUND

In programming, decision-making is one of the most important concepts. A computer program often needs to make decisions based on certain conditions. For example, a program may need to check whether a student has passed or failed, whether a user is eligible to vote, whether a number is positive or negative, or whether a password is correct.

Python provides conditional statements to perform decision-making. Conditional statements allow the program to execute different blocks of code depending on whether a condition is True or False.

A condition is usually written using comparison operators such as greater than, less than, equal to, not equal to, greater than or equal to, and less than or equal to.

Example:

Example:



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Python

marks = 75

Python

```
if marks >= 50:  
  
    print("Pass")
```

In this example, Python checks whether marks are greater than or equal to 50. If the condition is True, the message "Pass" is printed.

CONDITIONAL STATEMENTS IN PYTHON

Python mainly provides the following types of conditional statements:

1. if statement
2. if-else statement
3. if-elif-else statement
4. Nested if statement

THE IF STATEMENT

The if statement is used when we want to execute a block of code only if a condition is True.

Syntax:



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Python

```
if condition:  
    statement
```

Example:

Example:

Python

```
age = 18
```

Python

```
if age >= 18:  
    print("You are eligible to vote")
```

In this example, the condition `age >= 18` is True, so the print statement is executed.

THE IF-ELSE STATEMENT

The if-else statement is used when we have two possible outcomes. If the condition is True, the if block runs. If the condition is False, the else block runs.

Syntax:



Python

```
if condition:  
    statement  
  
else:  
    statement
```

Example:

Example:

Python

```
marks = 45
```

Python

```
if marks >= 50:  
    print("Pass")  
  
else:  
    print("Fail")
```

Here, the marks are less than 50, so the else block will run.

THE IF-ELIF-ELSE STATEMENT

The if-elif-else statement is used when we need to check multiple conditions.



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Syntax:

Python

```
if condition1:  
    statement  
  
elif condition2:  
    statement  
  
else:  
    statement
```

Python checks the conditions from top to bottom. As soon as one condition becomes True, that block is executed and the remaining conditions are skipped.

NESTED IF STATEMENT

A nested if statement means writing one if statement inside another if statement. It is used when one decision depends on another decision.

Syntax:

Python

```
if condition1:  
    if condition2:  
        statement
```



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```
else:  
    statement  
  
else:  
    statement
```

Example:

Example:

```
Python  
username = "admin"  
  
password = "12345"
```

8.

```
Python  
if username == "admin":  
    if password == "12345":  
        print("Login successful")  
    else:  
        print("Wrong password")  
else:
```



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```
print("Wrong username")
```

COMPARISON OPERATORS

Comparison operators are used to compare two values.

Comparison operators are used to compare two values.

Operator Meaning

== Equal to

!= Not equal to

Operator Meaning

== Equal to

!= Not equal to

> Greater than

Greater than

< Less than

< Less than

>= Greater than or equal to

<= Less than or equal to

= Greater than or equal to

<= Less than or equal to

Example:

Example:



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age >= 18
marks == 100
password != "12345"

**age >= 18
marks == 100
password != "12345"**

LOGICAL OPERATORS

Logical operators are used to combine multiple conditions.

Logical operators are used to combine multiple conditions.

Operator Meaning

and Returns True if both conditions are True
or Returns True if at least one condition is True
not Reverses the result of a condition

Operator Meaning

**and Returns True if both conditions are True
or Returns True if at least one condition is True
not Reverses the result of a condition**

Example:

Example:

marks >= 50 and attendance >= 75
age >= 18 or has_permission == True
not is_raining

**marks >= 50 and attendance >= 75
age >= 18 or has_permission == True
not is_raining**



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IMPORTANT NOTE ABOUT INDENTATION

Python uses indentation to identify blocks of code. Indentation means spaces at the beginning of a line. Usually, four spaces are used.

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Correct:

Correct:

```
if age >= 18:  
    print("Eligible")
```

```
if age >= 18:  
print("Eligible")
```

Incorrect:

Incorrect:

```
if age >= 18:  
print("Eligible")
```

```
if age >= 18:  
print("Eligible")
```

The incorrect code will generate an indentation error.

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PRACTICAL PROBLEMS WITH SOLUTIONS

PROBLEM 01: Check Voting Eligibility

Problem Statement:

Problem Statement:

Write a Python program that takes the age of a person as input and checks whether the person is eligible to vote or not.

Write a Python program that takes the age of a person as input and checks whether the person is eligible to vote or not.

Concept Used:

Concept Used:

if-else statement, Comparison operator, Input and type casting

if-else statement

Comparison operator

Input and type casting

Code:

Code:

Python

```
age = int(input("Enter your age: "))
```



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Python

```
if age >= 18:  
    print("You are eligible to vote.")  
else:  
    print("You are not eligible to vote.")
```

Expected Output:

Expected Output:

Enter your age: 20
You are eligible to vote.

Enter your age: 20
You are eligible to vote.

Explanation:

Explanation:

The program takes age as input from the user. Since input() returns data in string form, we convert it into an integer using int(). The condition age >= 18 checks whether the user is 18 years old or above. If the condition is True, the person is eligible to vote. Otherwise, the person is not eligible.

The program takes age as input from the user. Since input() returns data in string form, we convert it into an integer using int(). The condition age >= 18 checks whether the user is 18 years old or above. If the condition is True, the person is eligible to vote. Otherwise, the person is not eligible.

PROBLEM 02: Check Even or Odd Number

Problem Statement:



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Write a Python program that takes a number from the user and checks whether the number is even or odd.

Concept Used:

if-else statement

Modulus operator

Comparison operator

Code:

Python

```
number = int(input("Enter a number: "))
```

Python

```
if number % 2 == 0:  
    print("The number is even.")  
else:  
    print("The number is odd.")
```

Expected Output:

Enter a number: 7

The number is odd.

Explanation:

The modulus operator % gives the remainder after division. If a number is divided by 2 and the remainder is 0, then the number is even. Otherwise, it is odd.



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PROBLEM 03: Check Positive, Negative, or Zero

Problem Statement:

Write a Python program that takes a number from the user and checks whether it is positive, negative, or zero.

Concept Used:

if-elif-else statement
Comparison operators

Code:

Python

```
number = int(input("Enter a number: "))
```

Python

```
if number > 0:  
    print("The number is positive.")  
elif number < 0:  
    print("The number is negative.")  
else:  
    print("The number is zero.")
```

Expected Output:



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Enter a number: -5

The number is negative.

Explanation:

The program checks three possible conditions. If the number is greater than 0, it is positive. If the number is less than 0, it is negative. If both conditions are False, then the number must be zero.

PROBLEM 04: Student Grade System

Problem Statement:

Write a Python program that takes marks from the user and displays the grade according to the following criteria:

Marks 80 or above: Grade A

Marks 70 to 79: Grade B

Marks 60 to 69: Grade C

Marks 50 to 59: Grade D

Below 50: Fail

Concept Used:

if-elif-else statement

Comparison operators

Input and type casting

Code:

Python

```
marks = int(input("Enter your marks: "))
```



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Python

```
if marks >= 80:  
    print("Grade A")  
elif marks >= 70:  
    print("Grade B")  
elif marks >= 60:  
    print("Grade C")  
elif marks >= 50:  
    print("Grade D")  
else:  
    print("Fail")
```

Expected Output:

Enter your marks: 85

Grade A

Explanation:

Python checks the conditions from top to bottom. If marks are greater than or equal to 80, Grade A is printed. If not, Python checks the next condition. This process continues until one condition becomes True. If no condition is True, the else block runs.

PROBLEM 05: Find the Largest of Three Numbers

Problem Statement:



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Write a Python program that takes three numbers from the user and finds the largest number.

Concept Used:

if-elif-else statement

Logical operator and

Comparison operators

Code:

Python

```
a = int(input("Enter first number: "))  
b = int(input("Enter second number: "))  
c = int(input("Enter third number: "))
```

Python

```
if a >= b and a >= c:  
    print("The largest number is:", a)  
elif b >= a and b >= c:  
    print("The largest number is:", b)  
else:  
    print("The largest number is:", c)
```

Expected Output:



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Enter first number: 15

Enter second number: 25

Enter third number: 10

The largest number is: 25

Explanation:

The program compares three numbers. The first condition checks whether a is greater than or equal to both b and c. If this condition is True, a is the largest. Otherwise, the program checks b. If neither a nor b is the largest, then c is printed as the largest.

PROBLEM 06: Simple Login System Using Nested if

Problem Statement:

Write a Python program that takes username and password from the user. If the username is correct, then check the password. If both are correct, display "Login successful".

Correct username: admin

Correct password: 12345

Concept Used:

Nested if statement

String comparison

Input function

Code:

Python

```
username = input("Enter username: ")
```

```
password = input("Enter password: ")
```



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Python

```
if username == "admin":  
    if password == "12345":  
        print("Login successful.")  
    else:  
        print("Wrong password.")  
else:  
    print("Wrong username.")
```

Expected Output:

Enter username: admin

Enter password: 12345

Login successful.

Explanation:

First, the program checks the username. If the username is correct, then the program checks the password inside the first if block. This is called a nested if statement. If both username and password are correct, the login is successful.

PROBLEM 07: Exam Eligibility Checker

Problem Statement:

Write a Python program that checks whether a student is allowed to sit in the exam. A student is allowed if marks are greater than or equal to 50 and attendance is greater than or equal to 75.



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Concept Used:

if-else statement

Logical operator and

Comparison operators

Code:

Python

```
marks = int(input("Enter marks: "))  
  
attendance = int(input("Enter attendance percentage: "))
```

Python

```
if marks >= 50 and attendance >= 75:  
    print("You are allowed to sit in the exam.")  
else:  
    print("You are not allowed to sit in the exam.")
```

Expected Output:

Enter marks: 65

Enter attendance percentage: 80

You are allowed to sit in the exam.

Explanation:



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The program uses the logical operator and. Both conditions must be True. The student must have marks greater than or equal to 50 and attendance greater than or equal to 75. If any one condition is False, the student is not allowed.

PROBLEM 08: Simple Calculator Using if-elif-else

Problem Statement:

Write a Python program that takes two numbers and an operator from the user. The program should perform addition, subtraction, multiplication, or division based on the operator.

Concept Used:

if-elif-else statement

Arithmetic operators

Input and type casting

Code:

Python

```
num1 = float(input("Enter first number: "))  
num2 = float(input("Enter second number: "))  
operator = input("Enter operator (+, -, *, /): ")
```

Python

```
if operator == "+":  
    print("Result:", num1 + num2)
```



```
elif operator == "-":  
    print("Result:", num1 - num2)  
elif operator == "*":  
    print("Result:", num1 * num2)  
elif operator == "/":  
    if num2 != 0:  
        print("Result:", num1 / num2)  
    else:  
        print("Division by zero is not allowed.")  
else:  
    print("Invalid operator.")
```

Expected Output:

Enter first number: 10

Enter second number: 5

Enter operator (+, -, *, /): *

Result: 50.0

Explanation:

The program checks which operator the user entered. If the operator is +, addition is performed. If the operator is -, subtraction is performed. If the operator is *, multiplication is performed. If the operator is /, division is performed. A nested if is used inside division to check whether the second number is not zero.



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PROBLEM 09: Electricity Bill Calculator

Problem Statement:

Write a Python program that calculates the electricity bill according to the following conditions:

Units from 1 to 100: Rs. 10 per unit

Units from 101 to 200: Rs. 15 per unit

Units above 200: Rs. 20 per unit

Concept Used:

if-elif-else statement

Arithmetic operators

Comparison operators

Code:

Python

```
units = int(input("Enter electricity units: "))
```

Python

```
if units <= 100:  
    bill = units * 10  
  
elif units <= 200:  
    bill = units * 15  
  
else:
```



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```
bill = units * 20
```

```
print("Total electricity bill is Rs.", bill)
```

Expected Output:

Enter electricity units: 150

Total electricity bill is Rs. 2250

Explanation:

The program checks the number of units entered by the user. If units are less than or equal to 100, the bill is calculated using Rs. 10 per unit. If units are less than or equal to 200, Rs. 15 per unit is used. Otherwise, Rs. 20 per unit is used.

PROBLEM 10: Leap Year Checker

Problem Statement:

Write a Python program that checks whether a year is a leap year or not.

A year is a leap year if:

It is divisible by 4 and not divisible by 100, or

It is divisible by 400.

Concept Used:

if-else statement

Logical operators and/or

Modulus operator

Code:



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Python

```
year = int(input("Enter a year: "))
```

Python

```
if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):  
    print("The year is a leap year.")  
else:  
    print("The year is not a leap year.")
```

Expected Output:

Enter a year: 2024
The year is a leap year.

Explanation:

The program checks the rules of leap year. If the year is divisible by 4 and not divisible by 100, it is a leap year. A year is also a leap year if it is divisible by 400. The logical operators and or are used to combine these conditions.

COMMON ERRORS AND MISTAKES

1. Forgetting colon after if, elif, or else

Wrong:



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```
if age >= 18  
print("Eligible")
```

Correct:

```
if age >= 18:  
print("Eligible")
```

2. Wrong indentation

Wrong:

```
if marks >= 50:  
print("Pass")
```

Correct:

```
if marks >= 50:  
print("Pass")
```

3. Using = instead of ==

Wrong:

```
if password = "12345":  
print("Login")
```

Correct:

```
if password == "12345":  
print("Login")
```

4. Forgetting type casting with input

Wrong:

```
marks = input("Enter marks: ")
```



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```
if marks >= 50:  
    print("Pass")
```

Correct:

```
marks = int(input("Enter marks: "))
```

```
if marks >= 50:  
    print("Pass")
```

5. Writing conditions in the wrong order

Wrong:

```
if marks >= 50:  
    print("Pass")  
elif marks >= 80:  
    print("Grade A")
```

Correct:

```
if marks >= 80:  
    print("Grade A")  
elif marks >= 50:  
    print("Pass")
```

SUMMARY

Conditional statements are used to make decisions in Python programs.

The if statement runs a block of code only when a condition is True.

The if-else statement provides two possible paths.

The if-elif-else statement is used for multiple conditions.



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Nested if statements are used when one condition depends on another condition.

Comparison operators are used to compare values.

Logical operators are used to combine multiple conditions.

Proper indentation is very important in Python.

PRACTICE TASKS FOR STUDENTS

Note: Solve the following tasks by yourself. Do not copy from previous examples directly. Try to apply the concepts of if, if-else, if-elif-else, nested if, comparison operators, and logical operators.

Task 01:

Write a Python program that takes a temperature value from the user and checks whether the weather is cold, normal, or hot.

Task 02:

Write a Python program that takes a number from the user and checks whether it is divisible by both 3 and 5.

Task 03:

Write a Python program that takes marks of three subjects, calculates the average, and displays the grade.

Task 04:

Write a Python program that checks whether a person is eligible for a driving license. The person must be 18 or above and must have passed the driving test.

Task 05:



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Write a Python program that takes a product price and customer type. If the customer is a student, give 20% discount. If the customer is a regular customer, give 10% discount. Otherwise, no discount should be applied.